Server Setup Guide

Docker + Ollama installation and configuration on Ubuntu — pre-deployment reference

Contents

- 1. Install Docker Engine
- 2. Pull and Run the Ollama Container
- 3. Verify the Ollama Service
- 4. Pull and Run Language Models

> This guide documents the commands used to set up the server environment (Docker + Ollama) before pushing the project to GitHub. Run these on the target Ubuntu server via SSH.

1 Install Docker Engine

Update package lists and install the prerequisites needed to add Docker's APT repository.

```
sudo apt update
sudo apt install -y ca-certificates curl
```

Create the keyrings directory and download Docker's official GPG key.

```
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc
```

Register Docker's APT repository using the deb822 sources format.

```
sudo tee /etc/apt/sources.list.d/docker.sources <<EOF
Types: deb
URIs: https://download.docker.com/linux/ubuntu
Suites: $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}")
Components: stable
Architectures: $(dpkg --print-architecture)
Signed-By: /etc/apt/keyrings/docker.asc
EOF
```

Refresh package lists to pick up the new repository, then install Docker Engine and its plugins.

```
sudo apt update
sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
```

> This installs Docker CE, the CLI, containerd, the Buildx plugin, and the Compose plugin (`docker compose` v2 syntax).

2 Pull and Run the Ollama Container

Pull the official Ollama image and confirm it was downloaded.

```
sudo docker pull ollama/ollama
sudo docker images
```

Create a named Docker volume to persist model data across container restarts.

```
sudo docker volume create ollama
sudo docker volume ls
```

Run the Ollama container in detached mode, mapping the API port and mounting the persistent volume.

```
sudo docker run -d \
  --name ollama \
  -p 11434:11434 \
```

```
  -v ollama:/root/.ollama \
  ollama/ollama
```

Flag	Purpose
`-d`	Run container in the background (detached)
`--name ollama`	Assign a fixed container name for easy reference
`-p 11434:11434`	Expose the Ollama API on port 11434
`-v ollama:/root/.ollama`	Persist models/config in the `ollama` volume

> Port 11434 will be reachable on the server's network interface. If this is a public-facing server, restrict access with a firewall (e.g. `ufw`) or bind the port to localhost only.

3 Verify the Ollama Service

Confirm the container is running and that the API responds.

```
sudo docker ps
curl http://localhost:11434/api/tags
sudo docker volume ls
```

4 Pull and Run Language Models

Pull a model into the running container, then list the models available locally.

```
sudo docker exec -it ollama ollama pull qwen2.5-coder:7b
sudo docker exec -it ollama ollama list
```

Run the model interactively.

```
sudo docker exec -it ollama ollama run qwen2.5-coder:7b
```

Pull and run an additional, smaller model.

```
sudo docker exec -it ollama ollama pull gemma3:4b
sudo docker exec -it ollama ollama run gemma3:4b
```

> Models pulled with `ollama pull` persist in the `ollama` Docker volume, so they survive container restarts (but not volume deletion).